Two Dimensional Array

13 July 2026 16:45

A two dimensional (2D) array is an array whose elements are themselves arrays. It is commonly used to represent data in the form of rows and columns, just like a table, spreadsheet, chess board, sitting arrangement or matrix.

For example:

Roll no	Name	Marks
101	Amit	89
102	Priya	95
103	Rahul	78

This tabular data can be represented using a 2D array.

Declaration of a 2D Array
SyntaxL

```
dataType[][] arrayName;
      Or
dataType arrayName[][];
```

Example:

```
int[][] marks;
```

This statement only declares the array reference, no memory is allocated.

Creating a 2D Array
Memory is allocated using the **new** keyword.

Syntax:

```
Arrayname = new dataType[rows][columns];
```

Example:

```
marks = new dataType[3][4];
      Or
int[][] marks = new dataType[3][4];
```

This creates a 2D array with:

- **3 rows**
- **4 columns**
- **Total elements = 3 * 4 = 12**

Memory representation:

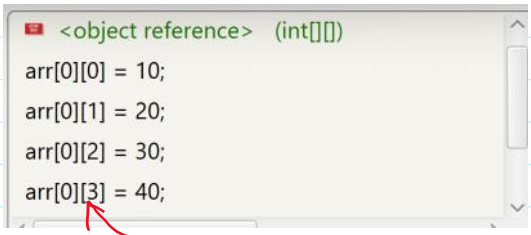
Row Column

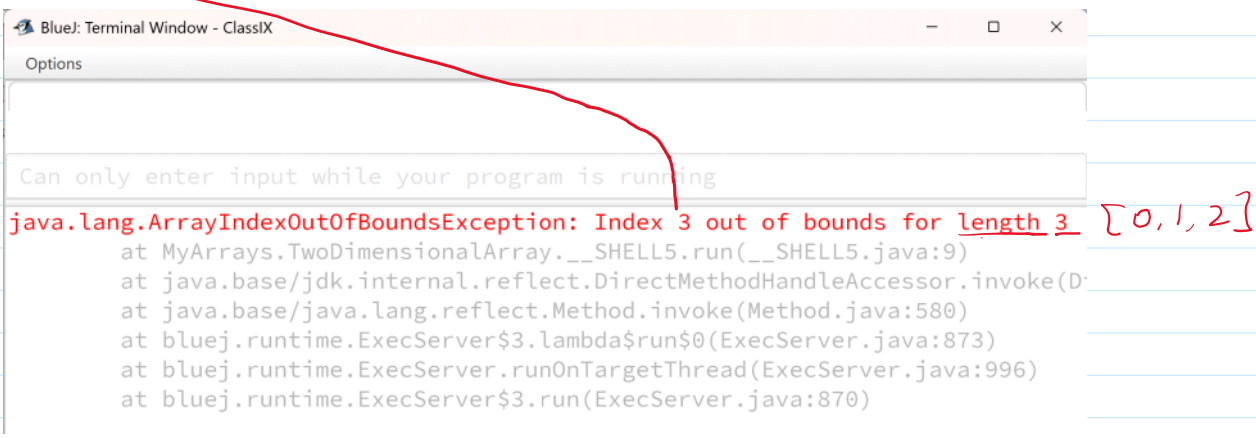
0,0	**0,1**	**0,2**	**0,3**
1,0	**1,1**	**1,2**	**1,3**
2,0	**2,1**	**2,2**	**2,3**

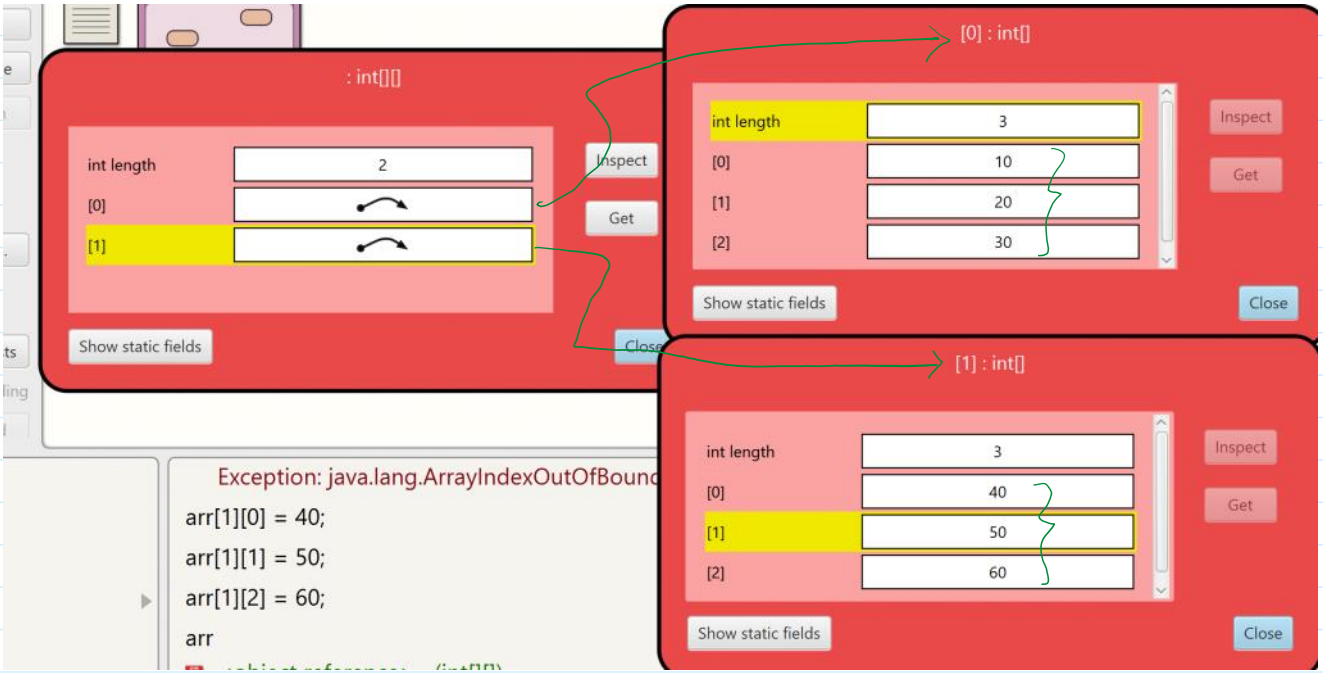
0
1
2

1,0	1,1	1,2	1,3
2,0	2,1	2,2	2,3

Initializing a 2D array
Method 1: Using Assignment





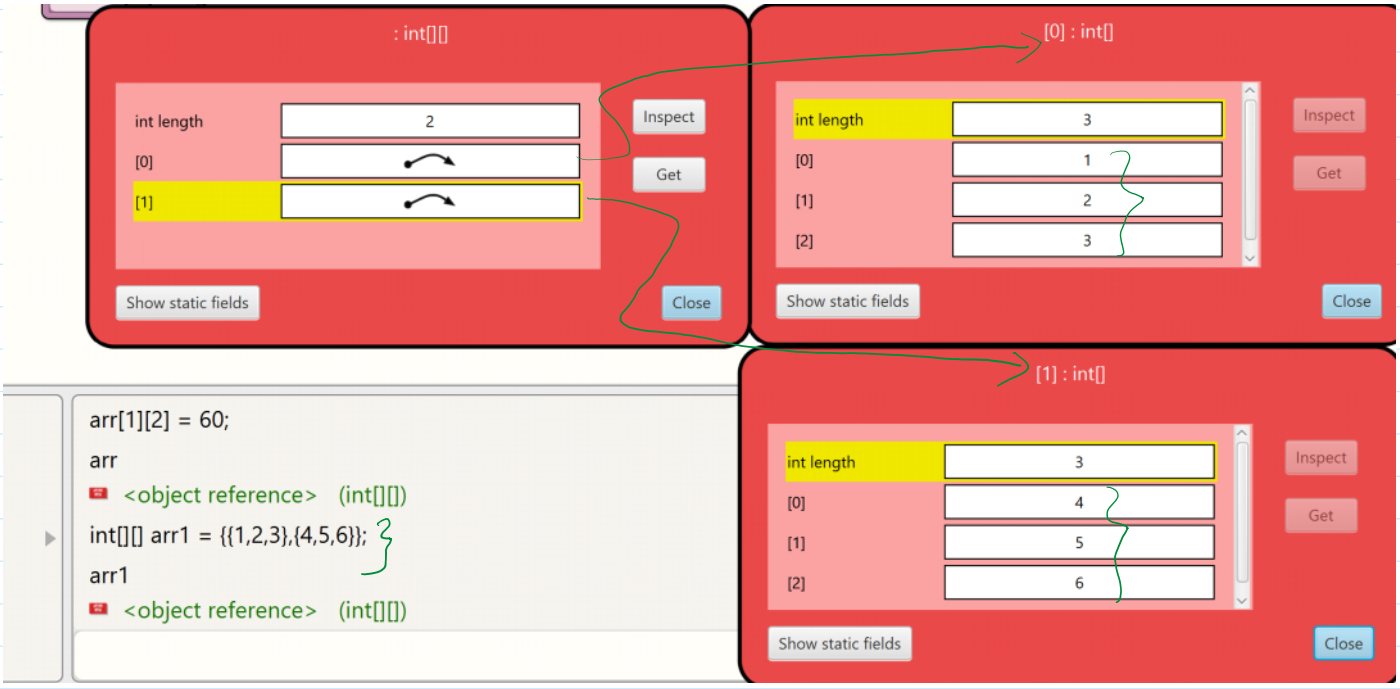


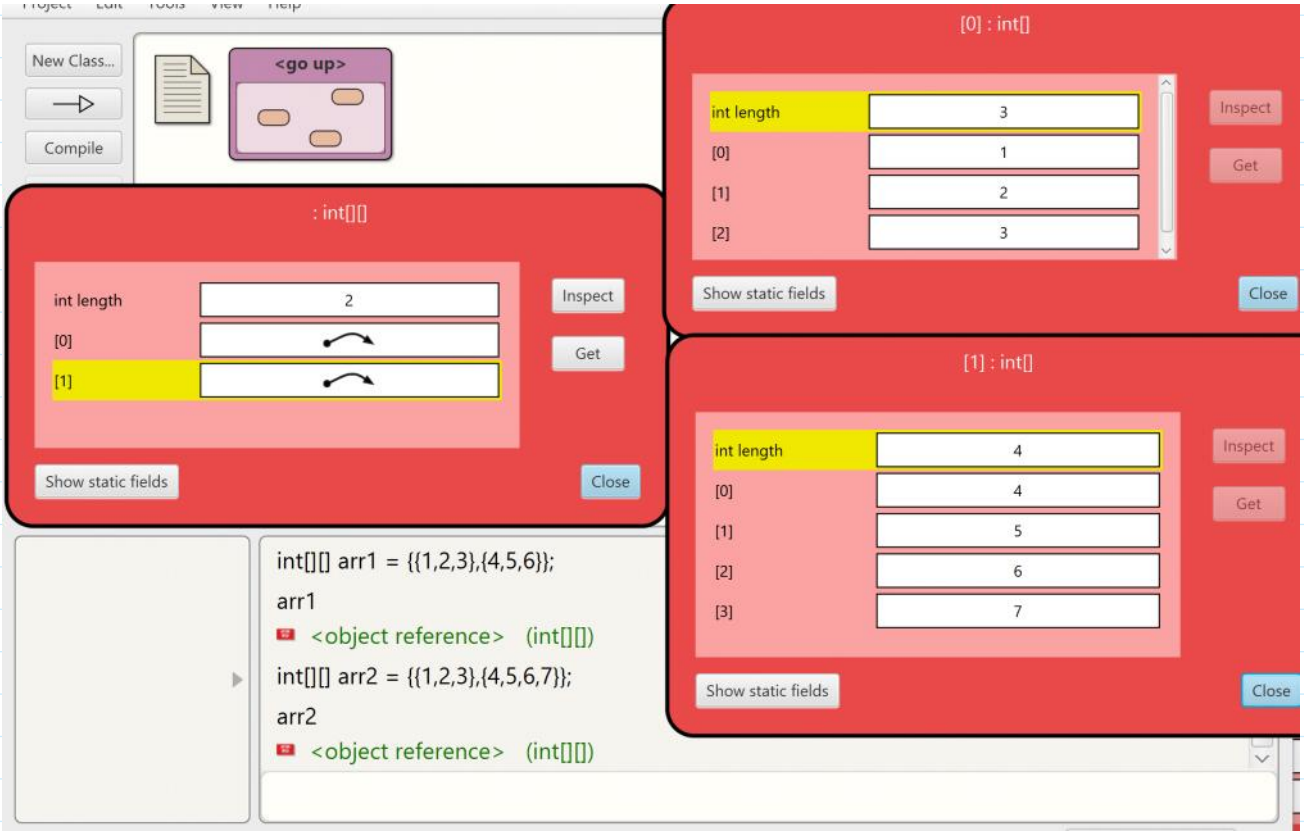

```
int[][] arr = new int[2][3];
arr
<object reference>   (int[][])
arr[0][0] = 10;
```

```
arr[0][1] = 20;
arr[0][2] = 30;
arr[0][3] = 40;
Exception: java.lang.ArrayIndexOutOfBoundsException (Index 3 out of bounds for
length 3)
arr[1][0] = 40;
arr[1][1] = 50;
arr[1][2] = 60;
arr
<object reference>   (int[][])
```

Method 2: Using Array Initializer

```
int[][] arr = {
   {10, 20, 30},
   {40, 50, 60}
};
```





Accessing Elements

Syntax:

```
arrayName[row][column];
```

Example:

```
System.out.print("\n"+arr2[0][2]);
```

Output:

```
3
```

Reading Values from the User

```
package MyArrays.TwoDimensionalArray;
import java.util.*;


/**
 * Write a description of class TwoDArrayUtil here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class TwoDArrayUtil
{
    private static Scanner sc = new Scanner(System.in);
    /**
     * This function takes data input in each element from the user.
     * @param arr int array in which data will inserted by the user
```

```
     * @param msg Any message or text displayed on the terminal.
     */
    public static void inputInt2DArray(int[][] arr, String msg){
        System.out.print("\n" + msg + "\n");
        int i, j;
        for(i = 0; i < arr.length; i++){
            for(j = 0; j < arr[i].length; j++){
                System.out.print("Enter integer value in ["+i+"]["+j+"]: ");
                arr[i][j] = sc.nextInt();
            }
        }
    }

    /**
     * This functions display on terminal int 2D array in matrix format
     * @param arr Two dimensional int array.
     * @param msg Any String message print with Array.
     */
    public static void printInt2DArray(int[][] arr, String msg){
        System.out.print("\n" + msg + "\n");
        int i, j;
        for(i = 0; i < arr.length; i++){
            for(j = 0; j < arr[i].length; j++){
                System.out.print("\t" + arr[i][j]);
            }
            System.out.print("\n");
        }
    }
}
```

Traversing of Array

```
package MyArrays.TwoDimensionalArray;
import java.util.*;


/**
 * Write a description of class Test2DArray here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Test2DArray
{
    public static void main(String[] args){
        int rows, columns;
        Scanner sc = new Scanner(System.in);
        System.out.print("\fEnter number of rows: ");
        rows = sc.nextInt();
        System.out.print("Enter number of columns: ");
        columns = sc.nextInt();
        int[][] A = new int[rows][columns];
        TwoDArrayUtil.inputInt2DArray(A, "Enter values in A[][]:");
```

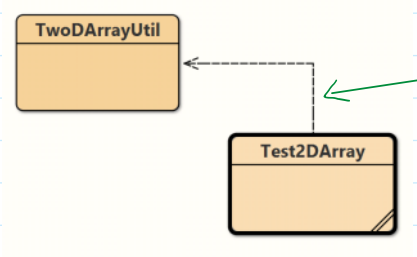
```
        TwoDArrayUtil.printInt2DArray(A, "Array A[][]: ");
    }
}
```

Output:

```
BlueJ: Terminal Window - ClassIX
Options

Enter values in A[][]:
Enter integer value in [0][0]: 10
Enter integer value in [0][1]: 12
Enter integer value in [0][2]: 14
Enter integer value in [1][0]: 16
Enter integer value in [1][1]: 18
Enter integer value in [1][2]: 20

Array A[][]:
        10      12      14
        16      18      20
```



Test2DArray class containing the reference of TwoDArrayUtil class. This concept is called Containership in OOP.